

“India is a really important market for a device like Raspberry Pi”



A credit-card sized computer with good capabilities at \$25 is definitely amazing. Many small-sized computers have come into the market in the past, but none made an impact like Raspberry Pi.

Diksha P. Gupta from Electronics For You caught up with Eben Upton, executive director, Raspberry Pi Foundation, to discuss the concept, price, market and everything else related to this computer. Read on...

EBEN UPTON
EXECUTIVE DIRECTOR, RASPBERRY PI

What made you think of a device like Raspberry Pi?

I was working for the University of Cambridge back in 2005-06. One of my jobs was to find high school students to come and study computer science at the university. A few of us noticed that the number of students opting for computer science was declining year after year, and their knowledge level was also declining simultaneously.

The students were bright but had less experience. So we had to spend more time to give them that experience after they joined the university. The challenge was to make them ready for the job or do a Ph.D., which requires a lot of experience. This generation is not familiar with the programming bit. The root cause is the rise of home PCs and gaming consoles that have replaced the programming machines we used in the 1980s. We used machines like BBC Micros, Amigas and Spectrum ZX, with which we learnt how to program. I used to program on my own because those machines tempted me into programming. I do not see much of that happening now. This is where we felt the need for a machine that was similar to what we used and which would prompt students to program on their own. We wanted to bring out a cost-friendly yet powerful device. That's how we came up with Raspberry Pi.

But why such a small device and why Linux?

Frankly, we were very driven by cost. Given that, we couldn't add too much onto the device and that allowed us to make it small. Also, making a small device reduces the cost because the PCB is smaller and cheaper. If the PCB is smaller, the manufacturing capacity also increases. So, if you could manufacture one large computer at a given time in a

panel, you can now make ten pieces of the Raspberry Pi in a panel. This reduces the manufacturing cost in a big way.

We wanted to make a product and driver with an emphasis on low cost. Linux was chosen for the same reason. Linux was the most cost-friendly option. If I was to design a \$400 PC, I could have justified a Windows licence costing \$50. But we were making a PC worth just \$25; we could have not justified that. More importantly, Linux addresses the issue of making a machine better. An operating system is to make the user understand how the machine works. With the availability of the source code, Linux serves the purpose better. Linux is free—both from cost and source code perspectives. This freedom is really important from the education standpoint.

Did you decide the cost of the device before you started designing Raspberry Pi?

Yes. We had some early prototypes ready but they didn't really meet our requirements. The feeling was that \$25 could be the price of a textbook. If you ask a student to purchase a special machine, those from the developed world may be able to afford a \$100 PC especially to learn programming. But it is certainly difficult for students from other parts of the world. So for those students, a \$25 device is absolutely apt and affordable.

Did you conduct any survey before you made Raspberry Pi?

It may sound crazy but we did not do any market research. We just thought of making a few computers that people may be interested in buying.